



Incident report analysis

Summary	A multimedia company that offers web design services, graphic design, and social media marketing solutions to small businesses recently experienced a DoS attack, which compromised the internal network for two hours until it was resolved. During the attack, the organization's network services suddenly stopped responding due to an incoming flood of ICMP packets. Normal internal network traffic could not access any network resources. The incident management team responded by blocking incoming ICMP packets, stopping all non-critical network services offline, and restoring critical network services. The company's cybersecurity team then investigated the security event. They found that a malicious actor had sent a flood of ICMP pings into the company's network through an unconfigured firewall. This vulnerability allowed the malicious attacker to overwhelm the company's network through a denial of service (DoS) attack.
Identify	Technology/Asset Management: "Which hardware devices, operating systems, and software were affected? Trace the flow of the attack through the internal network." Process/Business environment: "Which business processes were affected in the attack?"

	“Who needs access to the affected systems?” During the DoS attack , a firewall of the company’s network was unconfigured. This vulnerability allowed the threat actor to take advantage of the gap to use ICMP(internet Control Message Protocol) as a ping flood to overwhelm the company’s network. The organization’s network services suddenly stopped responding and normal internal network traffic could not access any network resources. Employees of the company experienced a 2 hour gap before resuming normal activity.
Protect	Develop and implement safeguards to protect the identified items and ensure delivery of services: ● Access control: Who needs access to the affected items? How are non-trusted sources blocked from having access? ● Awareness/Training: Who needs to be made aware of this attack and how to prevent it from happening again? ● Data security: Is there any affected data that needs to be made more secure? ● Information protection and procedures: Do any procedures need to be updated or added to protect data assets? ● Maintenance: Do any of the affected hardware, operating systems, or software need to be updated? ● Protective technology: Are there any protective technologies, like a firewall or an intrusion prevention system (IPS), that should be implemented to protect against future attacks?

	Only users within the network were affected. This time , the threat was not able to access the data but was able to use a firewall vulnerability. To address this security event, the network security team implemented: A new firewall rule to limit the rate of incoming ICMP packets Source IP address verification on the firewall to check for spoofed IP addresses on incoming ICMP packets Network monitoring software to detect abnormal traffic patterns An IDS/IPS system to filter out some ICMP traffic based on suspicious characteristics
Detect	Design and implement a system with tools needed for detecting threats and attacks: ● Anomalies and events: What tools could be used to detect and alert IT security staff of anomalies and security events, such as a security information and event management system (SIEM) tool? We could add a rule in a siem tool to detect early when an abnormal number of ping is detected. ● Security continuous monitoring: What tools or IT processes are needed to monitor the network for security events? We can use Splunk with Wireshark to monitor the network for security events ● Detection process: What tools are needed to detect security events, such as an IDS?

	In order to detect such attack quickly, we can use IDS and IPS (intrusion detection system / intrusion prevention system). An IDS will allow to get notified when intrusion occur and IPS will allow to stop the pings from coming if properly configured with a limit number of pings.
Respond	Design action plans for responding to threats and attacks: ● Response planning: What action plans need to be implemented to respond to similar attacks in the future? ● Communications: How will security event response procedures be communicated within the organization and with those directly affected by the attack, including end users and IT staff? ● Analysis: What analysis steps should be followed in response to a similar attack? ● Mitigation: What responding steps could be used to mitigate the impact of an attack, such as offlining or isolating affected resources? ● Improvements: What improvements are needed to improve response procedures in the future? The team must identify the source of where the ping flood is coming from and block the IP . Since it's a simple DoS attack , it can be stopped quickly for now. In the future , the company's has to have IDS and IPS working within the network to stop ICMP flood and other king of DoS attack as well as properly configured firewalls. In case of DDoS attack , blocking the IP sources will be inefficient.

Recover	Construct a plan and implement the framework for recovering and restoring affected systems and/or data: ● Recovery planning: How will resources be restored following an attack? ● Improvements: Do any improvements need to be made to the current recovery systems or processes? ● Communications: How will restoration procedures be communicated within the organization and with those directly affected by the attack, including end users and IT staff? This time , no hardware or software damage has been observed in the network's data and functions after resolving the DoS issue. Updating the firewall as well as adding IDS and IPS to the network will protect the company's network more effectively. This incident report analysis must be shared to the security team as well as maintenance team and another report that can be made public will have to be written if end users have been affected by this event.
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Reflections/Notes:

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